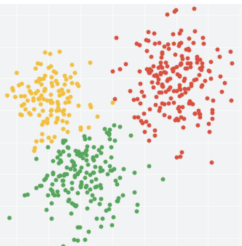


How to choose initial cluster representatives?

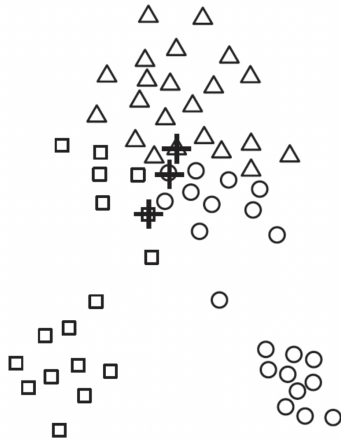
Procheta Sen



How to choose initial cluster representatives

1. Randomly
2. Randomly repeat several times
3. Sampling + hierarchical clustering
4. Furthest points
5. k -means++

Choosing randomly: lucky choice



Iteration 1

Choosing randomly: lucky choice

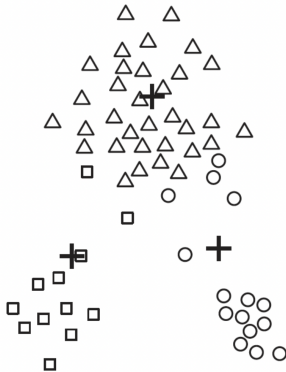


Figure: Iteration 2

Choosing randomly: lucky choice

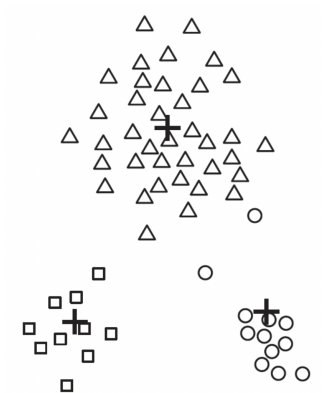


Figure: Iteration 3

Choosing randomly: lucky choice

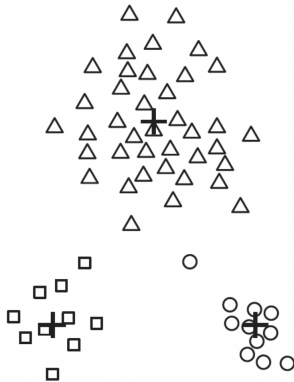


Figure: Iteration 4

Choosing randomly: unlucky choice

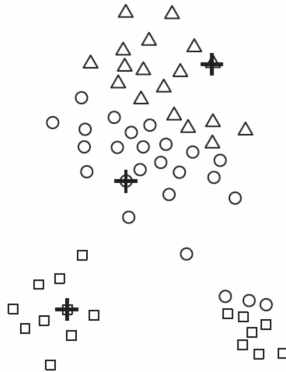


Figure: Iteration 1

Choosing randomly: unlucky choice

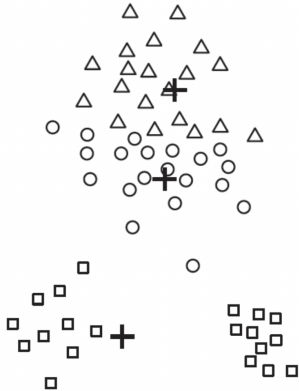


Figure: Iteration 2

Choosing randomly: unlucky choice

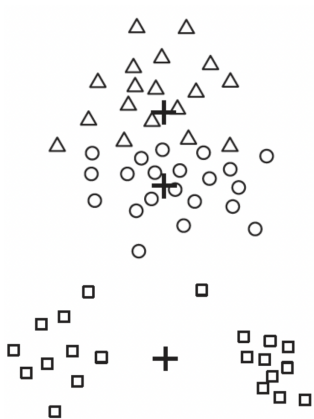


Figure: Iteration 3

Choosing randomly: unlucky choice

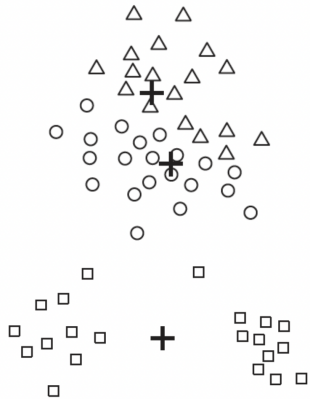
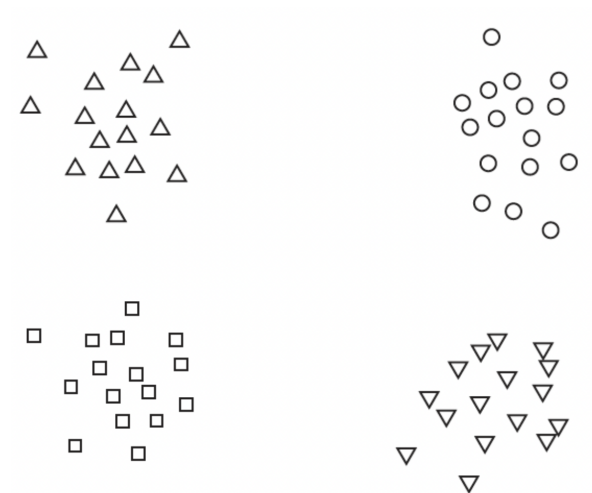


Figure: Iteration 4

Choosing randomly several times

1. Select initial cluster representatives randomly
2. Run k-means
3. Repeat steps 1–2 several times and choose the best clustering

Choosing randomly several times: lucky choice



Choosing randomly several times: lucky choice

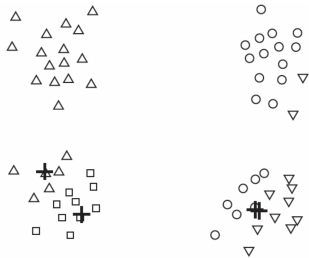


Figure: Iteration 1

Choosing randomly several times: lucky choice

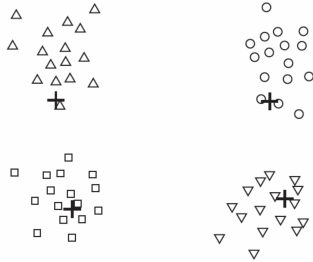


Figure: Iteration 2

Choosing randomly several times: lucky choice

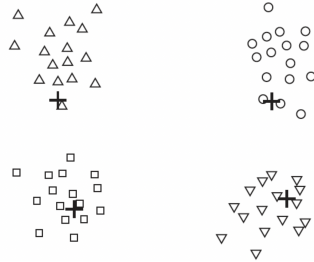


Figure: Iteration 3

Choosing randomly several times: unlucky choice

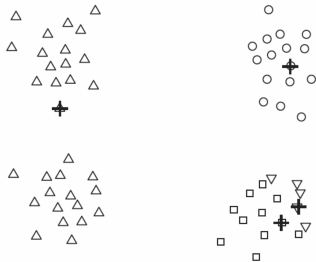


Figure: Iteration 1

Choosing randomly several times: unlucky choice

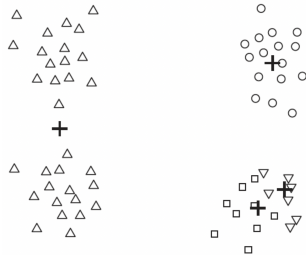


Figure: Iteration 2

Choosing randomly several times: unlucky choice

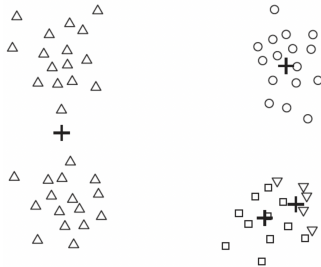


Figure: Iteration 3

Choosing randomly several times: unlucky choice

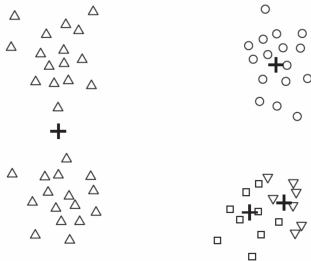


Figure: Iteration 4

Sampling + hierarchical clustering

1. Sample subset $\mathcal{D}'$ of points from the dataset $\mathcal{D}$
2. Cluster $\mathcal{D}'$ using a hierarchical clustering technique.
3. Extract k clusters from the hierarchical clustering.
4. Compute the means of these k clusters, and use them as the initial cluster representatives
5. Proceed with the standard k -means with these cluster representatives

Sampling + hierarchical clustering

Often works well, but it is practical only if

1. The sampled subset $\mathcal{D}'$ is relatively small (a few hundred to a few thousand), as hierarchical clustering is expensive
2. k is relatively small compared to the size of the sampled set $\mathcal{D}'$

Selecting furthest points

For an object $\bar{X}$ in the dataset $\mathcal{D}$ let $R(\bar{X})$ be the distance from $\bar{X}$ to the closest cluster representative we have already chosen.

1. Select one representative $\bar{Y}_1$ uniformly at random from $\mathcal{D}$.
2. For every $i = 2, \dots, k$
 - 2.1 Select representative $\bar{Y}_i$ from $\mathcal{D}$ with the maximum value of $R(\cdot)$
3. Proceed with the standard k-means using $\bar{Y}_1, \dots, \bar{Y}_k$ as initial cluster representatives

Selecting furthest points

The main drawback:

such an approach can select outliers, rather than points in clusters